

Chi-Squared, t , and F Distributions

1. The Chi-Squared Distribution

Definition

Let $Z_1, \dots, Z_k$ be independent standard normal random variables, that is,

$$Z_i \sim N(0, 1).$$

Then

$$Q = Z_1^2 + \dots + Z_k^2 \sim \chi_k^2.$$

The number k is called the number of *degrees of freedom*.

Standard example: the sample variance

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

and define

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The important sampling result is

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Why should $n-1$ degrees of freedom appear?

Start with a quantity whose distribution is immediate. Since

$$\frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

independently,

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi_n^2.$$

Now add and subtract $\bar{X}$:

$$X_i - \mu = (X_i - \bar{X}) + (\bar{X} - \mu).$$

Squaring and summing gives

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \bar{X}) + n(\bar{X} - \mu)^2.$$

But

$$\sum_{i=1}^n (X_i - \bar{X}) = 0,$$

so

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 = \frac{(n-1)S^2}{\sigma^2} + \frac{n(\bar{X} - \mu)^2}{\sigma^2}.$$

The left-hand side has a χ_n^2 distribution. Also,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1),$$

so

$$\frac{n(\bar{X} - \mu)^2}{\sigma^2} \sim \chi_1^2.$$

Thus the decomposition strongly suggests that the remaining term should have $n-1$ degrees of freedom.

Warning. This is only motivation. From an identity of the form

$$\chi_n^2 = A + \chi_1^2$$

one cannot simply conclude that $A \sim \chi_{n-1}^2$. But this is not yet a proof.

1. The Chi-Squared Distribution (continued)

The underlying theorem: orthogonal invariance

The actual reason comes from an important geometric property of the normal distribution.

Theorem (Orthogonal invariance of the standard normal). Let

$$Z = (Z_1, \dots, Z_n)^T,$$

where $Z_1, \dots, Z_n$ are independent $N(0, 1)$ random variables. If Q is any $n \times n$ orthogonal matrix, so that $Q^T Q = I$, then

$$QZ \stackrel{d}{=} Z.$$

In particular, the coordinates of QZ are again independent $N(0, 1)$ random variables.

In simple words: if we rotate a vector of independent standard normal random variables, the new coordinates are again independent standard normal random variables.

Applying the theorem to the sample variance

Standardize the sample:

$$Z_i = \frac{X_i - \mu}{\sigma}.$$

Choose an orthonormal coordinate system whose first unit vector is

$$u_1 = \frac{1}{\sqrt{n}}(1, \dots, 1).$$

After rotating coordinates, write the new coordinates as

$$Y_1, \dots, Y_n.$$

By the theorem, these are still independent $N(0, 1)$ random variables. The first coordinate is

$$Y_1 = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}.$$

Orthogonal transformations preserve squared length, so

$$\sum_{i=1}^n Z_i^2 = Y_1^2 + \dots + Y_n^2.$$

Combining this with the decomposition from the previous page gives

$$\frac{(n-1)S^2}{\sigma^2} = Y_2^2 + \dots + Y_n^2.$$

Since $Y_2, \dots, Y_n$ are independent standard normal random variables,

$$Y_2^2 + \dots + Y_n^2 \sim \chi_{n-1}^2.$$

Therefore

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

A useful geometric interpretation

The residuals satisfy

$$(X_1 - \bar{X}) + \dots + (X_n - \bar{X}) = 0.$$

So the residual vector lies in the $(n-1)$ -dimensional hyperplane

$$x_1 + \dots + x_n = 0.$$

Subtracting the sample mean removes exactly the single direction spanned by $(1, \dots, 1)$. This explains geometrically why one degree of freedom is lost.

One more consequence

In the rotated coordinates, $\bar{X}$ depends only on Y_1 , whereas S^2 depends only on $Y_2, \dots, Y_n$. Since these coordinates are independent,

For a normal sample, $\bar{X}$ and S^2 are independent.

This fact is exactly what is needed for the standard t -distribution example.

Remark. The decomposition above is a simple special case of *Cochran's theorem*, which gives a general method for splitting sums of squares of normal random variables into independent chi-squared parts. We will meet this theorem later in the context of ANOVA.

2. Student's t -Distribution

Definition

Let

$$Z \sim N(0, 1), \quad K \sim \chi_\nu^2,$$

and assume that Z and K are independent. Then

$$T = \frac{Z}{\sqrt{K/\nu}} \sim t_\nu.$$

The parameter ν is called the number of degrees of freedom. The t -distribution is symmetric about 0. It resembles the standard normal distribution, but has heavier tails. As $\nu \rightarrow \infty$, t_ν approaches $N(0, 1)$.

Standard example: estimating a mean when σ is unknown

Suppose

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2).$$

If σ were known, then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

In practice σ is often unknown, so we replace it by the sample standard deviation S . From the chi-squared result,

$$K = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

For a normal sample, $\bar{X}$ and S^2 are independent. Hence Z and K are independent, and therefore

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

What changed when σ was replaced by S ?

With known σ , standardization gives an exact standard normal variable. With unknown σ , we divide by the random quantity S instead. This additional randomness makes the tails heavier, producing a t -distribution rather than a standard normal distribution.

Important warning

The t -distribution conclusion does not follow merely from Z being standard normal and K being chi-squared. Their **independence is part of the definition** of the t -distribution. For normal samples, that independence ultimately comes from the same orthogonal-invariance argument used in the chi-squared discussion.

3. The F -Distribution

Definition

Let

$$U \sim \chi_r^2, \quad V \sim \chi_s^2,$$

and assume that U and V are independent. Then

$$F = \frac{U/r}{V/s} \sim F_{r,s}.$$

The numbers r and s are the numerator and denominator degrees of freedom. An F random variable is always nonnegative and its distribution is generally right-skewed.

Standard example: comparing two population variances

Suppose

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu_X, \sigma_X^2)$$

and

$$Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} N(\mu_Y, \sigma_Y^2),$$

with the two samples independent. Then

$$\frac{(n-1)S_X^2}{\sigma_X^2} \sim \chi_{n-1}^2, \quad \frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi_{m-1}^2,$$

and the two chi-squared random variables are independent. Hence

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{n-1, m-1}.$$

Equivalently,

$$\frac{S_X^2 \sigma_Y^2}{S_Y^2 \sigma_X^2} \sim F_{n-1, m-1}.$$

Why the null hypothesis is useful

Suppose we wish to test

$$H_0 : \sigma_X^2 = \sigma_Y^2.$$

Under H_0 , the unknown variance ratio cancels, and therefore

$$\frac{S_X^2}{S_Y^2} \sim F_{n-1, m-1} \quad \text{under } H_0.$$

Under the null hypothesis, the unknown variance ratio disappears. We are then left with a statistic that can be calculated from the data and whose distribution is known.

Useful facts and cautions

1. Independence of the numerator and denominator chi-squared variables is essential.
2. The exact variance-comparison result above assumes normal populations.
3. $F_{r,s}$ and $F_{s,r}$ are generally different distributions.
4. If $F \sim F_{r,s}$, then $1/F \sim F_{s,r}$.